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Debris filter for a nuclear fuel assembly bottom end part and method of manufacturing such a debris filter

Abstract

A debris filter configured for a nuclear fuel assembly bottom end part includes a lower nozzle (**8**) and the debris filter (**18**) is supported by the lower nozzle (**8**). The debris filter (**18**) has an inlet face (**18**A) and an outlet face (**18**B) opposed to the inlet face (**18**A), and comprises at least one filtering section (**18**D) that has a retention capacity that increases gradually or stepwise towards from the inlet face (**18**A) to the outlet face (**18**B).

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Background/Summary

(1) The present disclosure relates to nuclear fuel assemblies for use in nuclear reactors, in particular to nuclear fuel assemblies for use in a light water nuclear reactor.
BACKGROUND
(2) A nuclear fuel assembly for a light water reactor generally comprises a bundle of fuel rods extending along a longitudinal axis with being supported longitudinally and transversely in a spaced relationship by a support structure.
(3) In use, a plurality of nuclear fuel assemblies are positioned vertically and arranged side-by-side in a reactor vessel, thus forming a reactor core, and a coolant fluid, e.g. water, is caused to flow vertically through the nuclear fuel assemblies. The coolant fluid has the function of moderating the nuclear reaction and retrieving heat from the fuel rods of the nuclear fuel assemblies.
(4) Debris present in the coolant fluid may potentially damage the fuel rods of the nuclear fuel assembly and to hinder this risk, the nuclear fuel assembly may be provided with a bottom end part including a lower nozzle for receiving the coolant fluid flow and a debris filter, e.g. as disclosed in EP2204819A1 and EP2164076A1.
SUMMARY
(5) One of the aims of the present disclosure is to make it possible to provide a nuclear fuel assembly bottom end part debris filter that efficiently traps debris with low or adaptable pressure

losses, the debris filter being preferably easy to manufacture.
(6) To this end, the present disclosure provides a debris filter for a nuclear fuel assembly bottom end part comprising a lower nozzle and the debris filter supported by the lower nozzle, the debris filter having an inlet face and an outlet face opposed to the inlet face, wherein the debris filter comprises at least one filtering section that has a retention capacity that increases gradually or stepwise towards from the inlet face to the outlet face.
(7) With increasing retention capacity from the inlet face towards the outlet face, bigger debris are retained adjacent the inlet face and smaller debris penetrate into the debris filter but are retained adjacent the outlet face of the debris filter.
(8) A layer or portion of the debris filter with lower retention capacity and/or flow openings of larger dimensions can be made mechanically more resistant than a layer or portion of the debris filter with higher retention capacity and/or flow openings of smaller dimensions.
(9) The layer with higher retention capacity can be provided with walls thinner than that of the layer with lower retention capacity for maximizing the number of flow openings and limiting pressure losses through the debris filter.
(10) In specific embodiments, the debris filter comprises one or several of the following optional features, taken in isolation or in any technically feasible combination: the debris filter comprises several superimposed filtering layers located one above the other between the inlet face and the outlet face, the filtering layers having different retention capacities; each preceding filtering layer has a retention capacity strictly inferior to that of the next filtering layer when considering the filtering layers from the inlet face to the outlet face; it further comprising a plurality of filtering structures protruding on the inlet face of the debris filter, each filtering structure having a structure base and a structure apex spaced along a structure axis, each filtering structure comprising blades distributed circumferentially around the structure axis, each blade having one end connected to the structure base and one end connected to the structure apex, each blade delimiting a slot with each adjacent blade of the same filtering structure; each blade circumferentially overlaps each adjacent blade of the same filtering structure; each blade of each filtering structure extends helically around the structure axis of said filtering structure; each filtering structure is configured to impart a cyclone effect or a swirl around the structure axis to a fluid flowing through the filtering structure.
(11) The also relates to a nuclear fuel assembly comprising a bottom end part comprising a debris filter as defined above.
(12) The also relates to a method of manufacturing a debris filter as per any one of the preceding claims, comprising manufacturing the debris filter at least partly by additive manufacturing.
(13) In specific embodiments, the manufacturing method comprises one or several of the following optional features, taken in isolation or in any technically feasible combination: the debris filter is at least partly build up onto the lower nozzle by additive manufacturing; at least two of the filtering layers are manufactured separately and assembled together to obtain the debris filter; at least two filtering layers are made integrally in a single piece of material by additive manufacturing.

Description

BRIEF SUMMARY OF THE DRAWINGS
(1) The present disclosure and its advantages will be better understood upon reading the following description given solely by way of example and made with reference to the appended drawings, in which:
(2) FIG. **1** is a side elevation view of a BWR nuclear fuel assembly;
(3) FIG. **2** is a sectional view of a bottom end part of the nuclear fuel assembly;
(4) FIG. **3** is a sectional view of a bottom end part according to another embodiment;
(5) FIG. **4** is a partial side view of a debris filter;

(6) FIG. **5** is a sectional view of a filtering structure of the debris filter of FIG. **4**, taken along V-V on FIG. **4**:
(7) FIG. **6** is a partial top view of the debris filter of FIG. **4**;
(8) FIG. **7** is a side elevation view of a debris filter; and
(9) FIG. **8** is a side elevation view of a PWR nuclear fuel assembly.
DETAILED DESCRIPTION
(10) The nuclear fuel assembly **2** illustrated on FIG. **1** is elongated along a longitudinal axis L. In use, the nuclear fuel assembly is positioned in a nuclear reactor core with the longitudinal axis L extending substantially vertically, the nuclear fuel assembly resting on a lower core plate.
(11) In the following, unless otherwise specified, the terms “longitudinal”, “transversal”, “vertical”, “horizontal”, “top”, “bottom”, “upper” and “lower” are used with reference to the use position of the nuclear fuel assembly **2**.
(12) The nuclear fuel assembly **2** comprises a bundle of fuel rods **4** that are elongated and extend parallel to each other along the longitudinal axis L.
(13) Each fuel rod **4** comprises a tubular cladding receiving nuclear fuel, e.g. pellets of nuclear fuel stacked in the tubular cladding, and end caps closing the ends of the tubular cladding.
(14) The nuclear fuel assembly **2** comprises a support structure **6** supporting the fuel rods **4** longitudinally and transversely.
(15) The support structure **6** is configured to maintain the fuel rods **4** transversely in a spaced relationship. The fuel rods **4** are e.g. arranged in a lattice, at the nodes of an imaginary network, preferably a regular network. The transversal spacing between the fuel rods allows coolant fluid to flow between the fuel rods mainly longitudinally.
(16) The support structure **6** of the nuclear fuel assembly **2** comprises a lower nozzle **8**, an upper nozzle **10** and supporting grids **12** distributed along the fuel rods **4**. The fuel rods **4** extend between the lower nozzle **8** and the upper nozzle **10** with passing through the spacer grids **12**. The function of the spacer grids **12** is to maintain the fuel rods **4** longitudinally and transversely in the spaced relationship.
(17) The nuclear fuel assembly **2** of FIG. **1** is for use in a boiling water reactor (BWR).
(18) The armature **6** comprises a water channel **14** (or “water rods”) incorporated inside the bundle of fuel rods **4** by replacing fuel rods **4** in the lattice and a fuel channel **16** surrounding the bundle of fuel rods **4**.
(19) The water channel **14** and the fuel channel **16** extend between the lower nozzle **8** and the upper nozzle **10** with being connected to the lower nozzle **8** and the upper nozzle **10**. The spacer grids **12** are attached onto the water channel **14** are distributed at various positions along the water channel **14**.
(20) In use, the nuclear fuel assembly **2** is positioned vertically on a lower core plate with the lower nozzle **8** receiving a flow of coolant fluid which passes through the lower nozzle **8** and into to the bundle of fuel rods **4**, as illustrated by arrow F.
(21) The nuclear fuel assembly **2** comprises a debris filter **18** positioned inside the lower nozzle **8**. The lower nozzle **8** and the debris filter **18** define together a bottom end part **20** of the nuclear fuel assembly **2**.
(22) The debris filter **18** is configured for trapping debris present in the coolant fluid entering the nuclear fuel assembly **2** via the lower nozzle **8**.
(23) As illustrated on FIG. **2**, the lower nozzle **8** comprises a lower tie plate **22** and a nozzle part **24**.
(24) The lower tie plate **22** is configured notably for attaching the water channel **14** and the fuel channel **16** thereto. The nozzle part **24** is configured for receiving the coolant fluid flow and channeling it to the water channel **14** and the fuel channel **16**.
(25) The nozzle part **24** is tubular and extends downwardly from the periphery of the lower tie plate **22**. The nozzle part **24** comprises a lower inlet **26** and an upper outlet **28**.

(26) The lower tie plate **22** extends across the upper outlet **28** of the nozzle part **24**. The lower tie plate **22** has the shape of a grid. The lower tie plate **22** has fluid flow openings **30** for the coolant fluid to flow through the lower tie plate **22**.
(27) The nozzle part **24** has for example an upper section **32** of constant cross-section extending downwardly from the lower tie plate **22** and a lower section **34** of conical shape tapering downwardly.
(28) The nozzle part **24** defines an internal space **36**. The internal space **36** is delimited between the lower inlet **26** and the upper outlet **28** of the nozzle part **24**.
(29) The debris filter **18** is housed inside the lower nozzle **8**, more specifically here inside the nozzle part **24** and below the lower tie plate **22**. The debris filter **18** is located upstream the lower tie plate **22** when considering the coolant fluid flow in use. The debris filter **18** may be in contact with the underside of the lower tie plate **22**. Alternatively, a slight gap may be provided between the debris filter **18** and the underside of the lower tie plate.
(30) The debris filter **18** is preferably at least partly manufactured by additive manufacturing.
(31) Additive manufacturing refers to a manufacturing method in which the debris filter **18** or at least a part of the debris filter **18** is progressively manufactured by adding material to an already manufactured portion.
(32) Additive manufacturing refers in particular to progressively manufacturing at least a part of the debris filter by progressively adding manufacturing layers successively.
(33) The debris filter **18** is at least partly manufactured by additive manufacturing e.g. by using one of or a combination of the following additive manufacturing methods: powder bed additive manufacturing, in particular fusion, sintering or adhesive bonding), deposition additive manufacturing, in particular powder deposition additive manufacturing or wire deposition additive manufacturing, material Jetting additive manufacturing and binder jetting additive manufacturing.
(34) In one embodiment, the debris filter **18** is only partially manufactured by additive manufacturing.
(35) In one example, a first portion of the debris filter **18** is manufactured and a second portion of the debris filter **18** is built up onto the first portion by additive manufacturing.
(36) The first portion is preferably manufactured using a manufacturing method that is not additive manufacturing method. The first portion is e.g. manufacture by conventional milling of forged metal plates or casting.
(37) In another example, a first portion and a second portion of the debris filter **18** are manufactured separately, each of the first portion and the second portion being manufactured respectively by additive manufacturing, and the first portion and the second portion are then assembled together to obtain the debris filter **18**.
(38) In the above examples, the first portion is for example made of metal. The second portion is for example made of metal. The metal of the first portion and the metal of the second portion are the same metal or different metals.
(39) In one embodiment, the debris filter **18** is entirely manufactured by additive manufacturing. The debris filter is thus made integrally in a single piece of material obtained by additive manufacturing.
(40) In one embodiment, the debris filter **18** is manufactured separately from the lower nozzle **8** and is then assembled to the lower nozzle **8**.
(41) For example, a method of manufacturing the bottom end part **20** comprises manufacturing the nozzle part **24**, the lower tie plate **22** and the debris filter **18** separately and then assembling them.
(42) In one embodiment, the debris filter **18** is built up onto the lower nozzle **8** by additive manufacturing.
(43) The manufacturing method comprises for example manufacturing the lower tie plate **22**, then building up the debris filter **18** onto the lower tie plate **22** by additive manufacturing, and then assembling the nozzle part **24** to the lower tie plate **22**.

(44) Alternatively, the manufacturing method comprises for example manufacturing the nozzle part **24**, then building up the debris filter **18** onto the nozzle part by additive manufacturing, and then assembling the lower tie plate **22** to the nozzle part **24**.
(45) As illustrated on FIG. **2**, in one embodiment, the debris filter **18** extends along the longitudinal axis L over the majority of the extend E of the internal space **36** of the nozzle part **24** along the longitudinal axis L.
(46) Manufacturing the debris filter **18** at least in part by additive manufacturing allows providing a debris filter **18** that is thick to improve filtering efficiency whilst limiting pressure losses of the coolant fluid flowing through the debris filter **18**.
(47) The nozzle part **24** comprises here an upper portion **32** and a downwardly tapering lower portion **34**.
(48) In one embodiment, the debris filter **18** extends in the upper portion **32** of the nozzle part **24** at least down to the lower portion **34** of the nozzle part **24**.
(49) In one embodiment, the debris filter **18** extends in the upper portion **32** and in at least part of the lower portion **34** of the nozzle part **24**.
(50) In a specific embodiment, as illustrated on FIG. **2**, the debris filter **18** fully occupies the internal space **36** of the lower nozzle **8** between the lower inlet **26** and the upper outlet **28** of the nozzle part **24**.
(51) The debris filter **18** comprises an inlet face **18**A and an outlet face **18**B opposed to the inlet face **18**A. In use, the coolant fluid enters the debris filter **18** via the inlet face **18**A and exits the debris filter via the outlet face **18**B. The inlet face **18**A is adjacent the lower inlet **26** of the nozzle part **24** and the outlet face **18**B faces the lower tie plate **22**.
(52) The debris filter **18** is porous or has an openwork such as to allow coolant fluid to pass through the debris filter whilst retaining debris contained in the coolant fluid. The debris filter **18** comprises flow openings for allowing coolant fluid to flow through the debris filter **18** from the inlet face **18**A to the outlet face **18**B. Debris having dimensions greater than that of the flow openings will be retained by the debris filter **18**.
(53) The debris filter **18** has a retention capacity that corresponds to the size of the smallest debris that can be retained in the debris filter **18**. The higher the retention capacity, the smaller the size of the smallest debris retained by the debris filter **18**.
(54) The debris filter **18** will retain debris having dimensions larger than the dimensions of the flow openings. The smaller the dimensions of the flow openings, the higher the retention capacity of the debris filter **18**.
(55) In one embodiment, the debris filter **18** has a retention capacity that varies from the inlet face **18**A to the outlet face **18**B, and more specifically a retention capacity that progressively increases from the inlet face **18**A to the outlet face **18**B of the debris filter **18**.
(56) In view of progressively increasing the retention capacity through the thickness of the debris filter **18**, the debris filter **18** has for example flow openings that have dimensions that progressively diminish through the debris filter **18** from the inlet face **18**A to the outlet face **18**B of the debris filter **18**.
(57) With the progressively increasing retention capacity, the biggest debris are retained adjacent the inlet face **18**A of the debris filter **18** and smallest debris **18** penetrate into the debris filter **18** but are retained adjacent the outlet face of the debris filter **18**.
(58) One advantage of such configuration is that a layer or portion of the debris filter with lower retention capacity and/or flow openings of larger dimensions can be made mechanically more resistant than a layer or portion of the debris filter with higher retention capacity and/or flow openings of smaller dimensions. The layer with higher retention capacity can be provided with walls thinner than that of the layer with lower retention capacity for maximizing the number of flow openings and limiting pressure losses through the debris filter **18**.
(59) The additive manufacturing of the debris filter **18** allows tuning the retention capacity of the

debris filter **18** easily and notably providing a debris filter **18** with a retention capacity varying from the inlet face **18**A to the outlet face **18**B, in particular a retention capacity increasing gradually or stepwise from the inlet face **18**A to the outlet face **18**B, with limiting manufacturing cost and limiting pressure losses.
(60) The retention capacity of the debris filter **18** increases for example gradually or stepwise from the inlet face **18**A to the outlet face **18**B of the debris filter **18**. For example, the debris filter **18** has flow openings having dimensions that vary gradually or stepwise from the inlet face **18**A to the outlet face **18**B of the debris filter **18**.
(61) In the example illustrated on FIG. **2**, the debris filter **18** has a retention capacity that increases stepwise from the inlet face **18**A to the outlet face **18**B. For example, the debris filter **18** has flow openings having dimensions that vary stepwise from the inlet face **18**A to the outlet face **18**B.
(62) The debris filter **18** has for example several filtering layers **40**, **42**, **44** from the inlet face **18**A to the outlet face **18**B, each following layer having a higher retention capacity than the preceding layer when considering the filtering layers from the inlet face **18**A to the outlet face **18**B.
(63) Each filtering layer **40**, **42**, **44** has for example a constant retention capacity through the thickness of this filtering layer **40**, **42**, **44**.
(64) Each following filtering layer has e.g. flow openings of smaller dimensions than that of the preceding filtering layer when considering the filtering layers from the inlet face **18**A to the outlet face **18**B.
(65) The debris filter **18** thus has at least an inlet filtering layer **40** adjacent to the inlet face **18**A that has a lower retention capacity and an outlet filtering layer **44** adjacent to the outlet face **18**B of the debris filter **18**. The inlet filtering layer **40** has e.g. flow openings of larger dimensions than the flow openings of the outlet filtering layer **44**.
(66) In one embodiment, the debris filter has exactly two filtering layers, i.e. the inlet filtering layer **40** and the outlet filtering layer **44**.
(67) In another embodiment, the debris filter **18** has at least one intermediate filtering layer **42** interposed between the inlet filtering layer **40** and the outlet filtering layer **44**.
(68) In the example illustrated on FIG. **2**, the debris filter **18** has exactly one intermediate filtering layer **42**. In a variant, the debris filter **18** has more than two intermediate filtering layers and more than three filtering layers.
(69) In one embodiment of the manufacturing method, the filtering layers **40**, **42**, **44** of the debris filter **18** with different retention capacities are manufactured integrally in one piece of material by additive manufacturing. In this case, the debris filter **18** is manufactured integrally in one piece of material by additive manufacturing.
(70) In another embodiment, at least two filtering layers **40**, **42**, **44** are manufactured separately and then at least two filtering layers **40**, **42**, **44** are assembled together to obtain the debris filter **18**. The assembly is operated e.g. by welding, bonding, screwing, riveting . . . .
(71) As illustrated on FIG. **3**, in one embodiment, the debris filter **18** has a retention capacity that gradually increases.
(72) The debris filter **18** with gradually increasing retention capacity exhibits a gradient of retention capacity from the inlet face **18**A to the outlet face **18**B. The flow openings of the debris filter **18** exhibit e.g. a gradient of diminishing dimensions from the inlet face **18**A to the outlet face **18**B.
(73) The manufacture of the debris filter **18** by additive manufacturing allows imparting complex three-dimensional shapes to the debris filter **18** that may improve filtering whilst limiting pressure losses through the debris filter.
(74) FIGS. **4-6** illustrated one example of a debris filter **18** with a complex three-dimensional shape that can be obtained by additive manufacturing.
(75) As illustrated on FIGS. **4-6**, the debris filter **18** has an inlet face **18**A and an outlet face **18**B.
(76) The debris filter **18** is provided on its inlet face **18**A with a plurality of protruding filtering structures **50**. Each filtering structure **50** is configured for allowing coolant fluid to flow through

the filtering structure **50** while trapping the debris.

(77) Each filtering structure **50** is hollow. Each filtering structure **50** has a structure base **52** and a structure apex **54** spaced along a structure axis A and a plurality of blades **56** distributed circumferentially around the structure axis A with defining slots **58** between them. Each blade **56** has one end connected to the structure base **52** and one end connected to the structure apex **54**. Each blade **56** is located circumferentially between two adjacent blades **56** with defining a respective slot **58** with each of the two adjacent blades **56**.

(78) Each blade **56** has two lateral edges that are free. Each blade **56** extends like a bridge between its two ends connected respectively to the structure base **52** and the structure apex **54**.

(79) Each filtering structure **50** has a overall shape tapering from the structure base **52** to the structure apex **54**. The transversal dimensions of each filtering structure **50**, taken transversely to structure axis A, progressively diminish along the structure axis A, from the structure base **52** to the structure apex **54**. Each filtering structure **50** has for example an overall ogive shape.

(80) The structure axis A of the plurality of filtering structures **50** are preferably parallel. The structure axis A of each filtering structure **50** is preferably substantially perpendicular to the outlet face **18**B of the debris filter **18**. The structure axis A of each filtering structure **50** is preferably parallel to the longitudinal axis of the nuclear fuel assembly **2** when the debris filter **18** is mounted thereon.

(81) The structure bases **52** of the filtering structures **50** are connected together. The structure bases **52** define together an outlet portion **60** of the debris filter **18** defining the outlet face **18**B of the debris filter **18**.

(82) The filtering structures **50** impart a three-dimensional shaped to the inlet face **18**A of the debris filter **18**. The outlet face **18**A of the debris filter **18** is for example substantially flat. The outlet face **18**A of the debris filter is provided with outlet openings **62** for allowing the coolant fluid to exit the debris filter **18**.

(83) Each filtering structure **50** delimits a trapping volume delimited along the structure axis A by the structure base **52** and the structure apex **54** and delimited laterally by the blades **56**.

(84) The slots **58** delimited between the blades **56** allow the coolant fluid to enter inside the filtering structure **50** and the outlet openings **62** provided into the outlet face **18**B allow the coolant fluid to exit the filtering structure **50** and thus the debris filter **18**.

(85) The outlet face **18**B of the debris filter **18** exhibits e.g. a plurality of elongated outlet openings **62** extending in different transverse directions. Advantageously, as illustrated on FIG. **5**, the blades **56** of each filtering structure **50** are distributed around the structure axis A with a circumferential overlap between the adjacent blades **56**.

(86) Each blade **56** overlaps each adjacent blade **56** when looking in a direction that is radial relative to the structure axis A, as illustrated by arrow V on FIG. **5**. In other words, the slots **58** are not see-through when looking in a direction that is radial relative to the structure axis A.

(87) Preferably, each filtering structure **50** are configured to impart to the coolant flow flowing through the filtering structure **50** a cyclone effect or a swirl around the structure axis A of the filter structure **50**.

(88) Such a filtering structure **50** is particularly efficient to trap debris in the shape of small wires, which are very dangerous for the nuclear fuel assembly **2** because of the fretting mechanism. Owing to the cyclone effect, such debris are likely to be pressed into the filtering structure **50**, thus being prevented from exiting the filtering structure **50**.

(89) In one exemplary embodiment, the blades **56** of each filtering structure **50** are configured to generate the cycling effect with imparting to the coolant flow a cyclone effect or a swirl around the structure axis A of the filter structure **50**.

(90) As visible on FIG. **4**, advantageously, the blades **56** of each filtering structure **50** extend helically around the structure axis A. This improve trapping of elongated debris oriented longitudinally when reaching the inlet face of the debris filter **18**.

(91) The helical structure works like a cyclone and throw the debris outwards into the filtering structure **50**.
(92) In addition, such debris structure **50** tends to deviate the coolant flow with imparting transverse velocity to the coolant flow downstream the debris filter **18**. This can promote heat exchange between the coolant fluid and the nuclear fuel rods **4**.
(93) The debris filter **18** is very efficient for filtering debris, whilst limiting pressure loss. The debris filter has a complex shape but can be manufactured easily by additive manufacturing.
(94) At least the filtering structures **50** are manufactured by additive manufacturing. In one embodiment, the entire debris filter is manufacture by additive manufacturing. In another embodiment, the plate-shaped portion **60** is provided and then the remaining of the filtering structures **50** are built up onto the plate-shaped portion **50** by additive manufacturing.
(95) In one exemplary embodiment illustrated on FIG. **7**, a debris filter **18** may combine filtering structures **50** on the inlet face **18**A of the debris filter **18** as described in relation to FIGS. **4** and **5**, with a retention capacity increasing gradually or stepwise between the filtering structures **50** and the outlet face **18**B of the debris filter **18**.
(96) Such a debris filter **18** comprises for example an filtering inlet section **18**C defined by the filtering structures **50** protruding from the inlet face **18**A and an filtering outlet section **18**D extending between the filtering structures **50** and the outlet face with exhibiting a retention capacity increasing gradually or stepwise towards the outlet face **18**B.
(97) The outlet section **18**D comprises for example one or several filtering layer(s) located between inlet section **18**C the filtering structures **50** and the outlet face **18**B defining, e.g. one filtering layer of gradually increasing retention capacity or several superimposed filtering layers progressively increasing retention capacities, imparting a retention capacity increasing stepwise to the outlet section **18**D.
(98) In addition, a debris filter may have a filtering section exhibiting a retention capacity increasing gradually or stepwise associated with another filtering section located upstream or downstream a differing from the filtering structures **50**. In a general manner, a debris filter **18** may comprise at least filtering section exhibiting a retention capacity increasing gradually or stepwise from the side of the inlet face **18**A to the side of the outlet face **18**B.
(99) The present disclosure is not limited to the examples describes and illustrated above. Other examples may be contemplated.
(100) The debris filter **18** with specific debris designs described and illustrated above (increasing retention capacity and/or filtering structure with blades and/or cyclone effect) may be manufactured with other conventional manufacturing method not involving additive manufacturing. Besides, the present disclosure is not limited to BWR nuclear fuel assemblies. More generally, it is applicable to Light Water Reactor (LWR) nuclear fuel assemblies. In particular, it is applicable to Pressurized Water Reactor (PWR) nuclear fuel assemblies.
(101) FIG. **7**, in which numeral references to identical or similar elements to those of FIG. **1** are the same, illustrates a PWR nuclear fuel assembly.
(102) The nuclear fuel assembly comprises a bundle of fuel rods **4** extending along a longitudinal axis L and an armature **6** maintaining the fuel rods **4**.
(103) The armature **6** comprises a lower nozzle **8** and an upper nozzle **10** spaced along the longitudinal axis L, guide tubes **63** connected to the lower nozzle **8** and the upper nozzle **10** and spacer grids **12** distributed along the guide tubes **63** with being attached to the guide tubes **63**.
(104) The lower nozzle **8** is provided with a debris filter **18**. The lower nozzle **8** has an egg-crate plate **64** and feet **66** extending downwardly from corners of egg-crate plate **64**. In use, the nuclear fuel assembly **2** rests on the lower core plate via the feet **66**.
(105) The debris filter **18** is attached to the lower nozzle **8** below the egg-crate plate **64**. The debris filter **18** is thus located upstream the egg-crate plate **64**. The debris filter **18** is located between the feet **66**.

(106) The debris filter **18** may be as per the examples described with reference to the BWR nuclear fuel assembly.
(107) The method of manufacturing the debris filter **18** at least partly by additive manufacturing and the exemplary embodiments and variant described above (additive manufacturing onto the lower nozzle, layers obtained by additive manufacturing and assembled or integrally formed by additive manufacturing . . .) are advantageous independently from the shape of the debris filter.
(108) Hence, in a general manner, the present disclosure also relates to a method of manufacturing a debris filter for a bottom end part of a nuclear fuel assembly comprising a lower nozzle provided with the debris filter supported by the lower nozzle, the debris filter having an inlet face and an outlet face opposed to the inlet face, the method comprising manufacturing the debris filter at least partly by additive manufacturing.

Claims

1. A method of manufacturing a nuclear fuel assembly bottom end part comprising a lower nozzle and a debris filter supported by the lower nozzle, the lower nozzle comprising a first part and a second part, the debris filter having an inlet face and an outlet face opposed to the inlet face, the debris filter comprising at least one filtering section that has a retention capacity that increases gradually or stepwise towards from the inlet face to the outlet face, the method comprising: manufacturing the first part; manufacturing the second part; manufacturing the debris filter at least partly by building up the debris filter by additive manufacturing onto the first part; and then assembling the second part on the first part.
2. The method according to claim 1, wherein the first part is a nozzle part and the second part is a lower tie plate.
3. The method according to claim 1, wherein the debris filter comprises several superimposed filtering layers located one above another between the inlet face and the outlet face, the filtering layers having different retention capacities.
4. The method according to claim 3, wherein each preceding filtering layer has a retention capacity strictly inferior to that of the next filtering layer when considering the filtering layers from the inlet face to the outlet face.
5. The method according to claim 4, wherein at least two of the filtering layers are manufactured separately and assembled together to obtain the debris filter.
6. The method according to claim 4, wherein at least two filtering layers are made integrally in a single piece of material by additive manufacturing.
7. The method according to claim 1, wherein the second part is a nozzle part and the first part is a lower tie plate.